WrapEvoFS enables auditable feature compression with regret-constrained representative locking

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Abstract

Stochastic feature-selection pipelines can return signatures from the same data, yet choosing one run is often heuristic. We introduce WrapEvoFS, which first restricts selection to candidates within a prespecified empirical score gap and then locks the most representative eligible feature set. A corrected genetic objective also prevents zero truncation from erasing penalized-score ordering. In a fully nested four-class cancer analysis, all 30 locked signatures met the 0.01 score-gap tolerance, and 11 representative selections differed from the highest-locking-score candidate. Matched selector analyses showed branch-dependent trade-offs: WrapEvoFS had higher estimates than equally sized Elastic Net and stability rankings in one branch, similar estimates in another, and lower point estimates in the third, while native comparator supports were larger. A one-time post-freeze multicenter evaluation showed heterogeneous effects without reselection or retuning. Here, we show that stochastic feature compression can be made deterministic and empirically regret-bounded while exposing performance–compression and participant-partition sensitivity.

1 Introduction

High-dimensional biomedical tables often contain far more predictors than participants, making fitted models and selected signatures sensitive to preprocessing decisions, finite samples, and stochastic search paths [1–3]. Feature selection can make a model easier to inspect and deploy, but an apparently compact signature is not credible if held-out outcomes informed preprocessing, hyperparameters, a stochastic seed, or the reported feature set [4–7]. Favorable-run selection is a particularly quiet form of leakage: several stochastic runs are generated, but only the run with the most favorable downstream evaluation is reported.

RFECV provides an auditable recursive elimination path and can supply a target feature count, whereas genetic search can explore nonnested subsets but introduces run-to-run variability [8–11]. Stability selection and ensemble feature selection use resampling or aggregation to address selection variability [12, 13]; WrapEvoFS instead locks one representative member of a finite stochastic candidate bank after enforcing an explicit empirical score-gap constraint. Choosing only the highest-locking-score candidate preserves the best observed score but ignores whether the selected set is representative. Conversely, an unrestricted full-bank medoid may favor agreement at an unacceptably large development-score loss. The operational problem is therefore to select one representative stochastic output while making its empirical development-score gap explicit and bounded.

This problem is related to model multiplicity and Rashomon-set analysis, where multiple near-optimal models can differ in predictions or feature reliance [14, 15]. It also differs from multiobjective evolutionary feature selection, which searches performance–cardinality trade-offs during candidate generation [16]: WrapEvoFS applies a finite-bank feasibility constraint after candidate generation and then resolves multiplicity by representative selection. Nested selection bias and adaptive reuse of cross-validation scores remain separate concerns [4, 6, 17], as do algorithmic seed agreement and stability under participant resampling [18].

Audit of the original WrapEvoFS analyses identified a concrete objective-flattening mechanism: truncating a size-penalized genetic objective at zero made distinct negative penalized scores indistinguishable. The original algorithm remains available as a legacy-compatible mode, but this failure motivated a corrected untruncated objective that preserves penalized-objective ordering.

The study makes three contributions. First, it corrects objective flattening by using the untruncated objective for scientific ranking and run-best updating, with a nonnegative shift only for proportional parent-sampling weights. Second, it defines a finite-bank regret-constrained medoid: candidates within a prespecified empirical development-CV score-gap tolerance are eligible, and the eligible-pool Jaccard medoid is selected. Third, it evaluates the complete pipeline under fully nested resampling, adds matched selector-layer and random-bank controls, and tests frozen locks once across held-out centers. Deterministic content-derived tie-breaking and machine-readable artifacts make this hand-off reproducible without treating the hash or provenance record as a scientific score (Figure 1).

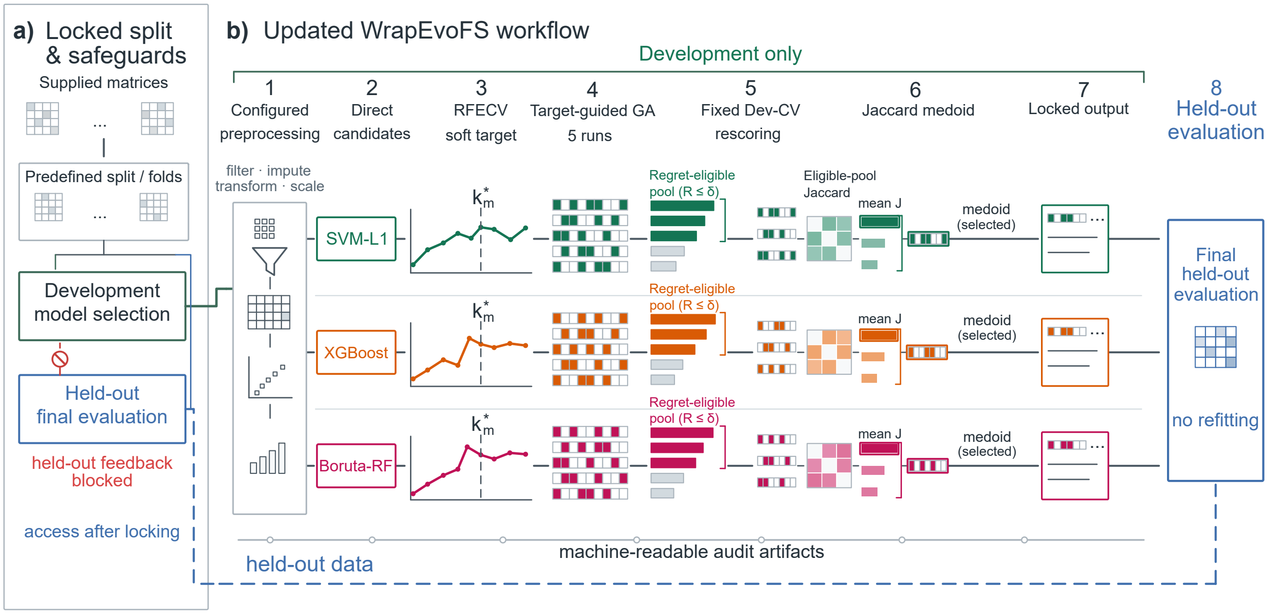

Figure 1. WrapEvoFS workflow and leakage boundary. a) Supplied matrices are assigned to predefined development and held-out partitions; development-only model selection blocks held-out feedback until locking. b) Configured preprocessing produces branch-specific Direct candidates, RFECV supplies a soft target, and five target-guided genetic searches generate retained candidates. Fixed development-CV rescoring defines empirical regret and the strict regret-eligible pool. Eligible-pool Jaccard similarity identifies the medoid, which becomes the locked output before final held-out evaluation. The blue boundary denotes held-out access after locking, and the lower trace denotes machine-readable audit artifacts. In the final block, "no refitting" means that held-out outcomes trigger no reselection or retuning; the prespecified final estimator is fitted on development data after locking.

We evaluate the corrected objective and locking rule under development-only, post-freeze held-out, and fully nested designs, while retaining archived analyses as explicitly labeled demonstrations.

Table 1. Empirical datasets, evaluation designs, outcome composition, and predictor inventories

Dataset	Endpoint	Evaluation design	Participants	Outcome composition	Input predictors
ADNI-derived multi-omics	CN / MCI / dementia	Fixed stratified 80:20 development/held-out split	665 total; 532 development; 133 held-out	Development 103/187/242; held-out 26/47/60	7,411 processed; 7,002 proteomic; 409 metabolomic
AMP-AD multicenter multi-omics	Ordered bScore: Low / Moderate / High	Four fixed leave-one-center-out folds	527; every participant held out once per prespecified configuration	111 / 156 / 260 overall; center-specific counts archived	13,136 supplied before fold filtering; 11,748 frontal proteomic; 1,388 metabolomic
CGGA all-glioma gene expression	MGMT status: methylated / unmethylated	Fixed stratified 70:30 development/held-out split	306 total; 214 development; 92 held-out	Development 110/104; held-out 47/45	24,326 gene-expression predictors
TCGA GBM/LGG multi-omics	Historical histological type: GBM / astrocytoma / oligodendroglioma / oligoastrocytoma	Two stratified five-fold outer-CV repeats	491; every participant held out once per repeat	66 / 147 / 165 / 113	25,118 supplied; 20,118 RNA-seq; 5,000 cleaned SCNV
VGH T1ce radiomics (H20-02354)	MGMT status: unmethylated / methylated	Fixed stratified 70:30 development/held-out split; two feature spaces share participants	197 total; 137 development; 60 held-out	Development 81/56; held-out 36/24	1,781 full-space; 1,346 label-independent perturbation-filtered

2 Results

2.1 Objective flattening and corrected-objective verification

Eighteen of 36 original AMP-AD configurations triggered at least one zero-truncation warning; none of six original CGGA configurations did. The most severe condition was AMP-AD Rush/SVM-L1/Small: target $k^* = 3$, locked count 84, absolute deviation 81, four of five zero-valued run-best fitnesses, and 245 all-zero generations. This is a concrete loss-of-ordering mechanism rather than a generic statement that genetic search is variable (Supplementary Fig. S1; Supplementary Tables S1–S2).

The matched development-only comparison comprised 120 completed GA runs in 24 Small/Reference center–branch–cap conditions (Figure 2; Supplementary Fig. S2; Supplementary Tables S1–S2). Aggregate absolute target deviation decreased from 216 to 137; the corrected-objective configuration was better in 12 conditions, unchanged in 6, and worse in 6. The prespecified Rush/SVM-L1/Small stress condition accounted for 56 of the 79 reduced deviation units; excluding it, deviation remained lower at 135 versus 112. All-zero generations decreased from 673 to 333 overall and from 428 to 267 without the stress condition.

All 24 regret-constrained medoids satisfied the prespecified absolute empirical regret tolerance of 0.01; the maximum selected regret was 0.00835 (Figure 2c). Corrected-objective-minus-original fixed locking-score differences had mean +0.00098, median +0.00096, and range −0.01319 to +0.02389; uniform parent-sampling fallbacks were zero. Candidate generation and locking both changed, so this comparison verifies configuration-level target fidelity and score-gap behavior rather than isolating either component. Held-out outcomes were accessed only after the protocol and all 24 selected masks were frozen.

2.2 Regret-constrained locking enforces finite-bank feasibility

The four locking rules answer distinct decision questions (Table 2). The highest-locking-score candidate guarantees zero empirical regret but does not optimize representativeness. The unrestricted full-bank medoid optimizes full-bank representativeness without a nontrivial score-gap guarantee, whereas the legacy top-three medoid substitutes a rank cutoff for a metric-scale constraint. The regret-constrained medoid first enforces eligibility and only then optimizes within-pool mean Jaccard.

Table 2. Decision rules for selecting one feature set from a finite stochastic candidate bank.

Rule	Admissible pool	Primary decision	Empirical-regret guarantee	Representativeness target
Highest-locking-score candidate	Full bank	Maximize locking score	Zero	Not explicitly optimized
Legacy top-three medoid	Three highest-ranked candidates	Maximize mean Jaccard within the rank-restricted pool	No metric-scale bound	Rank-restricted pool
Unrestricted full-bank medoid	Full bank	Maximize mean Jaccard within the full bank	No nontrivial bound	Full candidate bank
Regret-constrained medoid	$E_\delta = \{r : R_r = L_{\text{best}} - L_r \leq \delta\}$	Maximize mean Jaccard within E_δ	$R_{\text{selected}} \leq \delta$	Strict regret-eligible pool

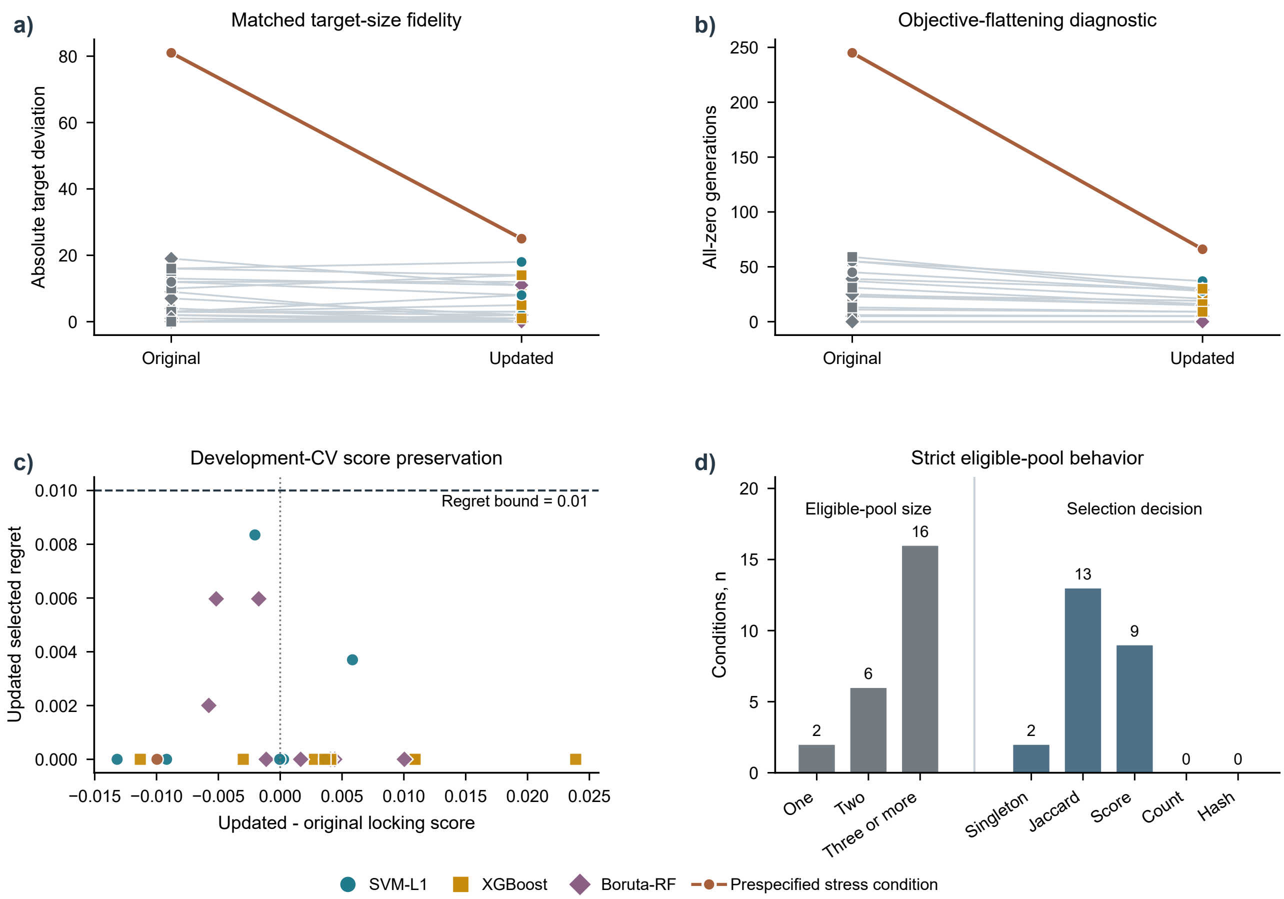

Figure 2. Development-only verification of the corrected-objective WrapEvoFS configuration. a) Condition-level original-versus-corrected absolute target deviation and b) all-zero-generation diagnostics across 24 center–branch–cap conditions and 120 full GA runs. Gray denotes original observations, branch colors denote corrected-objective observations, and the highlighted pair denotes the prespecified Rush/SVM-L1/Small stress condition. Aggregate target deviation was 216 versus 137 overall and 135 versus 112 after excluding the stress condition; corresponding all-zero-generation totals were 673 versus 333 and 428 versus 267. c) Corrected-minus-original fixed development-CV locking-score difference versus selected empirical regret; marker shape identifies branch and the dashed line marks the 0.01 bound. d) Eligible-pool size and final selection-decision stage; numerical labels are condition counts. No held-out outcomes were accessed.

The existing-artifact 2 by 2 cross-lock analysis separated candidate-bank and locking-rule effects (Supplementary Table S3). Within 24 original-objective banks, the regret-constrained medoid changed 10 masks and reduced mean and maximum selected regret from 0.00261 and 0.02111 to 0.00060 and 0.00729, while total target deviation changed from 216 to 227. Within the 24 corrected-objective banks, it changed 8 masks and reduced mean and maximum regret from 0.00351 and 0.01525 to 0.00108 and 0.00835, while target deviation changed from 135 to 137. The rule therefore bounds configured score gaps without optimizing target fidelity.

At $\delta = 0.01$, 2 of the 24 corrected-objective conditions had singleton eligible pools, 6 had two candidates, and 16 had at least three (Figure 2d; Supplementary Table S4). Unique Jaccard medoids determined 13 selections, higher locking score resolved 9, and singleton selection resolved 2; feature count and stable hash were not reached. No exact duplicate mask occurred in the 48 complete banks, and all recovered Rush selections matched their archived and frozen-evaluation records.

Across 36 archived AMP-AD banks, retrospective regret-constrained re-locking compressed the corresponding Direct sets by a median 74.1% (range 15.8%–80.0%); median empirical score gap was 0 and the maximum was 0.0091. Across six CGGA banks, median compression was 57.9% (range 34.3%–70.0%), median gap was 0.00065, and maximum gap was 0.00355 (Supplementary Fig. S3; Supplementary Table S5). Recovery of the complete five-mask ADNI banks enabled the same audit without rerunning selection or accessing held-out outcomes. At absolute $\delta = 0.01$, all three eligible pools were singletons and the selected empirical regret was zero. The rule retained the legacy XGBoost selection but changed the SVM-L1 and Boruta-RF selections, for which the legacy score gaps were 0.0140 and 0.0225, respectively (Supplementary Table S8).

2.3 Tolerance sensitivity and seed agreement

Increasing absolute tolerance enlarged the strict eligible pool in AMP-AD and CGGA (Supplementary Fig. S4), and all 168 selections met their declared threshold without pool expansion. We fixed $\delta = 0.01$ before locking. This absolute tolerance is defined on the reported metric scale and is not a universal threshold. Complete five-mask matrices yielded median selected mean Jaccard values of 0.204 and 0.424, respectively; median Nogueira coefficients were negative and 0.076. These values quantify stochastic agreement on fixed development samples, not participant-resampling or biomarker stability.

2.4 Fully nested benchmark reveals selector-dependent trade-offs

A separate TCGA GBM/LGG experiment moved the complete selection pipeline inside two repeated five-fold outer-CV partitions for 491 participants. The design yielded 150 candidates in 30 banks without outer-test access during selection or locking. Every selected candidate satisfied the prespecified 0.01 empirical-regret tolerance; the maximum was 0.00978. The regret-constrained medoid differed from the canonical highest-locking-score candidate in 11 banks, and every change increased eligible-pool mean Jaccard (median gain 0.0429; Figure 4a,b; Supplementary Fig. S5).

Mean within-bank pairwise Jaccard among the five GA candidates was 0.115, whereas mean Jaccard among locked signatures across outer folds was 0.0357 (Figure 4c). Across outer partitions, similar signature size did not imply similar feature identity: 42 of 60 paired locked signatures had cardinality similarity at least 0.9, yet 33 of those 42 pairs had feature-set Jaccard at most 0.05 (Figure 4d). Supplementary Fig. S6 reports chance-corrected candidate agreement, all dependent outer-fold pairwise similarities, and RNA/SCNV composition. These are descriptive measures of stochastic-search agreement and participant-partition sensitivity within this cohort, not estimates of biomarker stability.

The regret-constrained-medoid signatures contained a mean of 14.6, 20.0, and 18.0 features in the SVM-L1, XGBoost, and Boruta-RF branches, respectively, compared with 81, 100, and 100 Direct features. Mean repeated out-of-fold (OOF) macro one-vs-rest AUROC was 0.827, 0.827, and 0.828. Corresponding WrapEvoFS-minus-RFECV-only differences were −0.0108 (95% CI −0.0259 to 0.0035), 0.0012 (−0.0113 to 0.0134), and 0.0054 (−0.0125 to 0.0225); all intervals included zero (Figure 3a).

A protocol-frozen retrospective selector-layer analysis compared these signatures with nested multinomial Elastic Net, subsampling stability selection, and five independently seeded random banks within each identical post-Direct outer-training universe (Figure 3b–d; Supplementary Figs. S7–S8; Supplementary Table S6). At matched cardinality, SVM-L1 WrapEvoFS exceeded Elastic Net by 0.0176 (95% CI 0.0008 to 0.0341) and stability selection by 0.0261 (0.0082 to 0.0434). XGBoost differences were −0.0017 (−0.0146 to 0.0111) and −0.0015 (−0.0136 to 0.0105). Boruta-RF differences were −0.0109 (−0.0236 to 0.0024) and −0.0128 (−0.0260 to 0.0012). Native Elastic Net and stability supports used mean branch-specific counts of 60–77 and 31–58 features, whereas WrapEvoFS used 15–20. No selector dominated all branches on both performance and feature count. WrapEvoFS provides a compact, traceable hand-off subject to the regret constraint; predictive performance remained branch dependent.

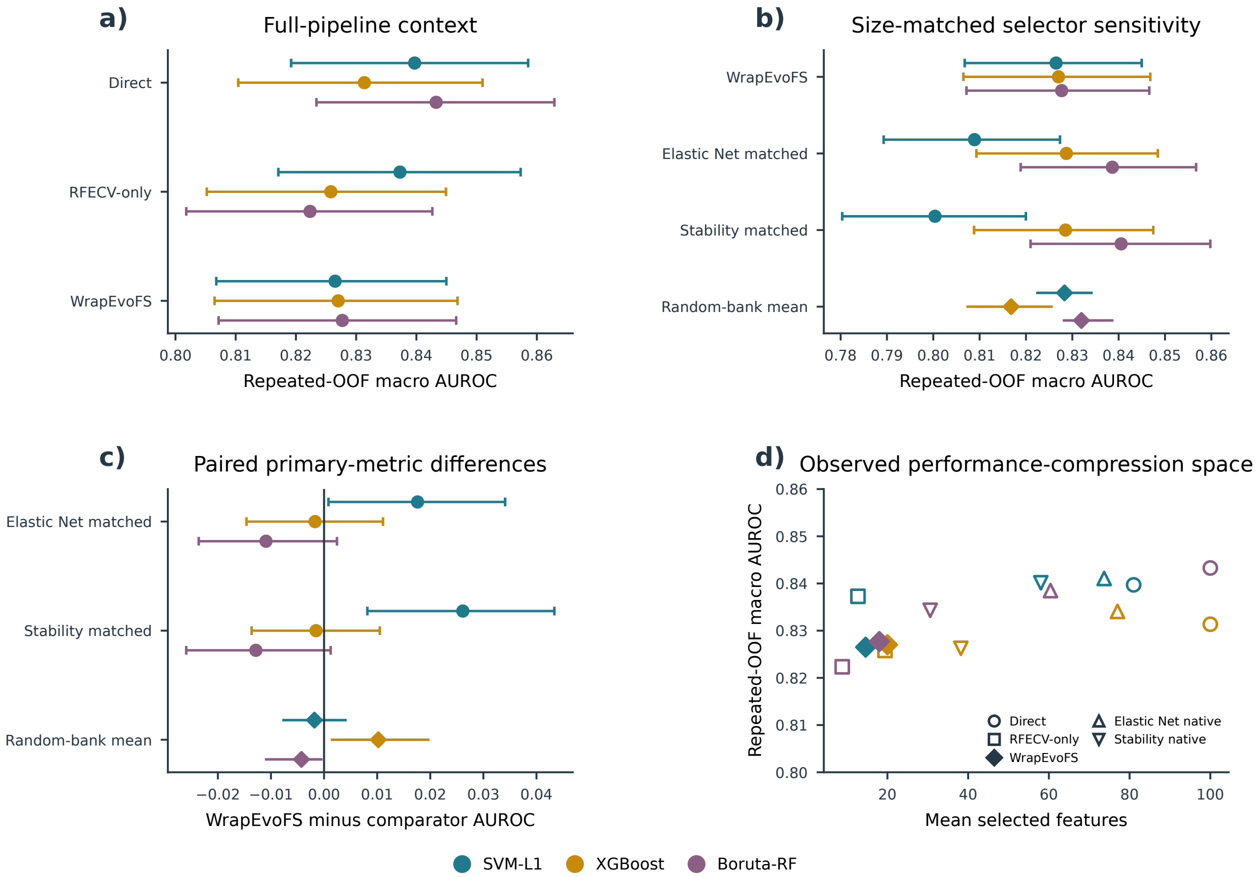

Figure 3. Fully nested TCGA performance–compression benchmark. All estimates use the same 491 participants, two stratified five-fold outer-CV repeats, training-only selection, and a fixed final random forest. Colors identify Direct branches. a) Repeated-OOF macro one-vs-rest AUROC for Direct, RFECV-only, and regret-constrained-medoid WrapEvoFS signatures. b) AUROC for WrapEvoFS, cardinality-matched Elastic Net and stability rankings, and five random-bank replicates summarized by their mean diamond and range. c) Paired WrapEvoFS-minus-comparator AUROC differences; intervals for Elastic Net and stability selection use 2,000 participant-clustered class-stratified resamples, while random-bank symbols show the range across five prespecified replicate banks. d) Mean feature count versus AUROC for Direct, RFECV-only, WrapEvoFS, and native Elastic Net and stability supports. The selector-layer comparisons are conditional on the same post-Direct universe and do not establish standalone pipeline equivalence.

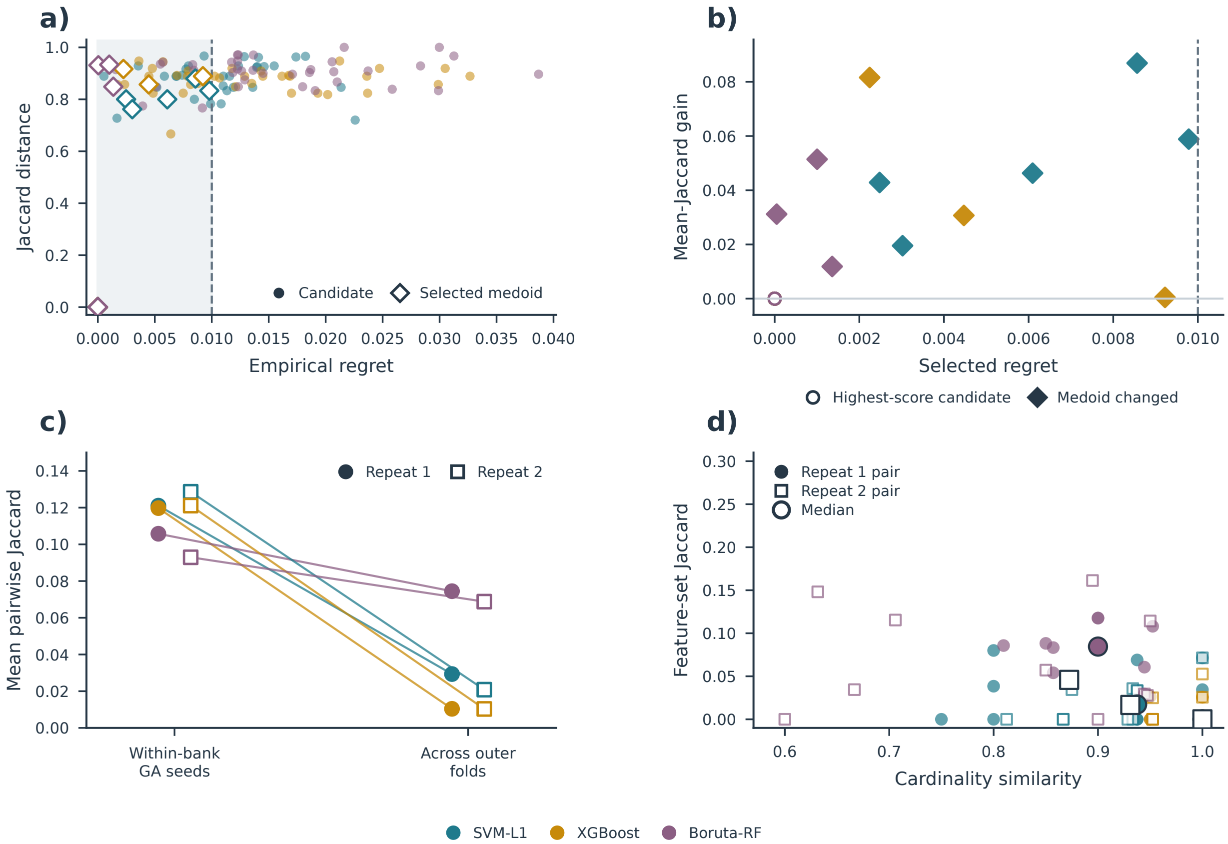

Figure 4. Fully nested analysis of regret-constrained locking geometry in TCGA GBM/LGG. The design comprised 491 participants, two stratified five-fold outer-CV repeats, three Direct branches, and five saved GA candidates per outer-training bank, yielding 150 candidates in 30 banks. Outer-test records were not used for feature selection or locking. a) Empirical development-CV regret and Jaccard distance from the canonical highest-locking-score mask for all candidates. Shading and the dashed line denote the absolute regret-eligible region and $\delta = 0.01$; open diamonds identify selected medoids. Highest-locking-score candidates overlap at regret and distance zero. b) Mean eligible-pool Jaccard gain of the selected medoid over the highest-locking-score candidate in 24 nonsingleton pools. Filled diamonds mark 11 changed selections; open circles mark retained highest-locking-score candidates. Singleton pools are omitted because the gain is undefined. c) Mean pairwise Jaccard among candidates within each bank and among locked signatures across outer folds, summarized by repeat and branch. d) Feature-count similarity, defined as $\min(k_1, k_2)/\max(k_1, k_2)$, versus feature-set Jaccard for all 60 outer-fold pairs; large symbols show coordinate-wise branch-repeat medians. All selected empirical regrets were at most 0.01. The agreement estimates characterize candidate-search and participant-partition sensitivity within this cohort.

2.5 Frozen selections undergo one-time multicenter evaluation

After the 24 corrected-objective AMP-AD locks and the evaluation protocol were frozen, each signature was evaluated once in its corresponding held-out center without reselection or parameter changes (Figure 5; Supplementary Fig. S9; Supplementary Table S7). Pooled differences ranged from -0.0404 to $+0.0480$ against RFECV-only and from -0.0311 to $+0.0176$ against the legacy top-three medoid. Four of six and five of six 95% intervals included zero, respectively, and the two non-zero-crossing intervals favored different methods. Because the masks and evaluation protocol were frozen before held-out outcome access, this analysis evaluates protocol adherence and center transport. The heterogeneous effects do not support a uniform predictive claim.

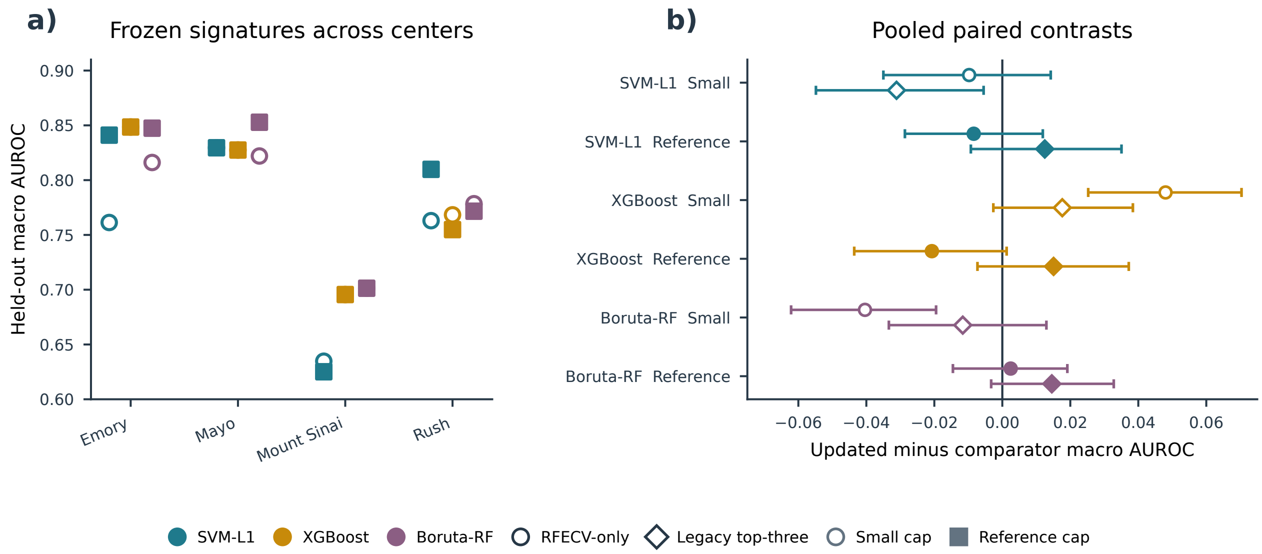

Figure 5. One-time post-freeze AMP-AD held-out evaluation. a) Macro one-vs-rest AUROC for 24 frozen center–branch–cap signatures; colors identify branches, open circles denote the Small cap, and filled squares denote the Reference cap. b) Pooled regret-constrained-medoid-minus-RFECV-only circles and regret-constrained-medoid-minus-legacy-top-three-medoid diamonds with 95% intervals from 2,000 common resamples stratified by held-out center and true class (seed 42). Open and filled symbols denote Small and Reference caps. All masks, parameters, and the protocol hash were frozen before held-out outcomes were loaded; evaluation triggered no reselection or retuning.

2.6 Archived and cross-modal analyses define empirical scope

Archived original-configuration demonstrations remain fully reported in the Supplement. ADNI locked signatures reduced Direct counts by 44%–70%, but all nine primary paired held-out intervals crossed zero (Supplementary Figs. S10–S12; Supplementary Tables S8–S9). Original AMP-AD effects varied by center, branch, cap, metric, and comparator (Supplementary Figs. S13–S14; Supplementary Tables S10–S12). CGGA showed 54%–73% compression and variable five-run agreement without a uniform predictive advantage in its coherent same-split benchmark (Supplementary Figs. S15–S19; Supplementary Tables S13–S18).

The VGH brain-tumour radiomics cohort (UBC REB H20-02354, “Brain Tumor Image Analysis”) provided a supplementary internal test in a distinct feature modality. The same 197 MGMT-linked participants were analyzed under a frozen 137/60 development/held-out split in full and label-independent perturbation-filtered feature spaces (Supplementary Fig. S20). All six locks had zero empirical development-CV regret at $\delta = 0.01$, with 10.0%–81.0% compression relative to Direct sets (median 75.6%). Paired held-out locked-minus-Direct AUROC estimates ranged from -0.007 to 0.069, and all 95% intervals included zero (Supplementary Table S19). This analysis broadens the tested modality but reuses participants across feature spaces, relies on unadjudicated automatic regions, and remains internal.

2.7 Secondary held-out trade-offs and interoperability

Secondary tuned-model, computational, random-subset, and Elastic Net analyses are reported in Supplementary Figs. S21–S23 and Supplementary Table S16. Bayesian, STABL, and BLiP analyses used the locked ADNI outputs as reproducible downstream candidate sets without reopening held-out-informed selection (Supplementary Figs. S24–S27; Supplementary Tables S20–S22). Five of eight features selected by training-only STABL overlapped the 78-feature WrapEvoFS locked-signature union, yielding an 81-feature candidate set for restricted downstream analyses (Supplementary Figs. S26–S27; Supplementary Table S21). The corresponding 81-feature restricted Laplace Bayesian model yielded a held-out macro one-vs-rest AUROC of 0.803 (95% CI 0.744–0.856), compared descriptively with 0.780 for the 7,411-feature Bayesian reference, for which a confidence interval was unavailable (Supplementary Figs. S24–S25 and S27; Supplementary Tables S20–S22). Directions varied by method, and agreement is not independent replication because the analyses reuse the same development boundary.

2.8 Controlled candidate-bank locking simulation

A prespecified simulation isolated the locking layer under hidden utility across 4,995,000 synthetic banks (Supplementary Fig. S28; Supplementary Table S23). In the primary $R = 5$, $\delta = 0.01$ design, the regret-constrained medoid had no empirical-regret violation and increased mean full-bank Jaccard from 0.424 to 0.446 relative to highest-locking-score selection, but mean hidden-oracle regret was 0.00259 versus 0.00222. It combined lower oracle regret with higher representativeness in 91 of 162 primary scenarios, with relative behavior depending on score noise, consensus alignment, bank size, and tolerance. The simulation quantifies the feasibility–representativeness trade-off; it does not show uniformly lower oracle regret.

3 Discussion

WrapEvoFS separates stochastic candidate generation from the decision to retain one feature set. Its corrected objective preserves penalized-objective ordering that zero truncation can erase, and regret-constrained representative locking imposes an explicit empirical development-CV score-gap constraint before selecting the eligible-pool medoid. The fully nested benchmark adds a practical result: no tested selector dominated every Direct branch on both primary predictive performance and feature count. WrapEvoFS retained 15–20 features; compared with equally sized Elastic Net and stability rankings, it had higher estimates in the SVM-L1 branch, similar estimates in the XGBoost branch, and lower point estimates with zero-crossing intervals in the Boruta-RF branch. The post-freeze AMP-AD analysis further demonstrated that masks and parameters can be fixed before held-out-center outcomes are accessed.

The comparator results clarify when the additional stochastic search and locking layer may be useful. Native Elastic Net and stability supports generally achieved higher AUROC than their matched-cardinality versions but required substantially larger signatures. At equal cardinality, the direction depended on the upstream Direct universe, and random-bank means likewise varied by branch. Pipeline comparison and locking-rule comparison therefore answer different questions: RFECV, Elastic Net, and stability selection generate feature sets, whereas the highest-locking-score candidate, legacy top-three medoid, unrestricted full-bank medoid, and regret-constrained medoid choose among a supplied stochastic bank. The proposed rule is most relevant when nonnested candidate exploration is scientifically justified and one compact, traceable output must be retained.

Five saved seeds reveal dispersion across run-best masks, while the nested TCGA experiment shows a distinct participant-partition effect: similar selected counts can coexist with markedly different feature identities across outer partitions (Figure 4c,d; Supplementary Fig. S6). The nested analysis therefore separates cardinality stability from feature-identity stability. Representative locking standardizes the candidate passed downstream, but it does not resolve sensitivity to participant composition. Seed agreement and participant-resampling stability should therefore be reported as different estimands [12, 13, 18].

The formal guarantee is finite-bank and empirical: it concerns only the configured development-CV score gap. In archived and fixed-partition analyses, preprocessing and Direct screening were fitted on each full development partition rather than within locking-score folds, and repeated fold reuse makes locking scores adaptive internal selection scores [17]. Five seeds are sparse. TCGA nested the complete pipeline, but the new selector comparison is conditional on post-Direct universes, remains internal, and was protocol-frozen retrospectively after the parent TCGA results existed. The radiomics analysis lacks scanner/site covariates, repeat segmentations, and manual ROI adjudication, and post-freeze center effects were heterogeneous. Broader cohorts, participant-resampling designs, and biological follow-up are needed to establish transportability and biomarker reproducibility; these boundaries do not alter the proved score-gap guarantee or reproducible selection audit.

In summary, WrapEvoFS offers an auditable rule for selecting one representative subset from a stochastic candidate bank under a strict empirical score-gap constraint. WrapEvoFS standardizes selection from a stochastic candidate bank under an explicit empirical development-CV score-gap tolerance and reports signature-size and feature-identity stability separately.

4 Methods

4.1 Overall analytical workflow and leakage boundary

Figure 1 separates development-only selection from one-time held-out evaluation. Preprocessing and Direct screening precede RFECV target estimation; five genetic searches per branch generate retained run-best subsets that a common development-CV evaluator rescores. A prespecified tolerance defines the regret-eligible pool, whose Jaccard medoid is locked under deterministic tie-breaking. Held-out outcomes cannot determine the strategy, tolerance, objective, metric, tie rule, or selected run, and trigger no reselection. Complete pseudocode and executable settings are provided in the Supplementary Methods and Supplementary Table S24.

4.2 RFECV target and primary estimands

For branch m, Direct denotes the branch-specific pre-GA screening procedure (SVM-L1, XGBoost, or Boruta-RF) [9, 19, 20], and its candidate set is $\mathcal{C}_m$ with size p_m. RFECV evaluates its recursive elimination path using development folds and chooses the smallest count attaining the largest mean score among counts allowed by the configured cap:

$$k_m^* = \min\left(\arg\max_{k\in\mathcal{K}_m^{\mathrm{elig}}} \overline{S}_{R,m}(k)\right). \tag{1}$$

For selected set F, compression relative to the Direct set is

$$C(F) = 1 - \frac{|F|}{p_m}. \tag{2}$$

All locking metrics are oriented so that larger values are better. Every retained candidate is evaluated on the same K_L fixed stratified folds. Let $s_{r\ell}^L$ be the locking-metric score for candidate r on fold ℓ, obtained by fitting the fixed-specification random forest [21] on that fold's training rows using feature set F_r and the candidate's recorded estimator seed η_r, then scoring its validation rows. The locking score is simply the mean of these fold scores:

$$L_r = \frac{1}{K_L}\sum_{\ell=1}^{K_L} s_{r\ell}^L. \tag{3}$$

All candidates within a bank share the same fold partition, metric, and estimator hyperparameters; run-specific estimator seeds are recorded as part of deterministic candidate rescoring. The search-stage metric defined below may differ from M_L, because search proposes candidates whereas the common locking evaluator compares them. With $L_{\mathrm{best}} = \max_r L_r$, empirical development-CV regret is

$$R_r = L_{\mathrm{best}} - L_r. \tag{4}$$

Run-to-run agreement is summarized by mean pairwise Jaccard, selected-candidate mean Jaccard, and the chance-corrected Nogueira coefficient only when a complete run-by-feature matrix exists [18]. Because each condition has five stochastic seeds on one fixed development sample, these are seed- or stochastic-agreement measures, not participant-resampling or biomarker stability. Held-out performance differences are secondary and are computed only after locking:

$$\Delta_{\mathrm{heldout}} = M(F^{\mathrm{lock}}) - M(\mathcal{C}_m). \tag{5}$$

4.3 Legacy-compatible and corrected genetic objectives

A nonempty binary chromosome $\mathbf{z} \in \{0,1\}^{p_m}$ represents feature set $F_m(\mathbf{z})$. The GA uses K_G fixed stratified folds and a configured search-stage metric. Let $s_{m\ell}^G(\mathbf{z};\xi)$ be the metric score for chromosome $\mathbf{z}$ on fold ℓ, obtained with the configured random-forest evaluator and seed $\xi + \ell$, where ξ is the chromosome-level seed. Its base score is the mean of the fold scores:

$$\overline{S}_{G,m}(\mathbf{z};\xi) = \frac{1}{K_G}\sum_{\ell=0}^{K_G-1} s_{m\ell}^G(\mathbf{z};\xi). \tag{6}$$

The folds evaluate a frozen development-derived candidate matrix; upstream preprocessing and Direct screening are not refitted inside these scoring folds. Let λ be the penalty per feature of deviation from k_m^*. Both modes compute

$$Q_m(\mathbf{z};\xi) = \overline{S}_{G,m}(\mathbf{z};\xi) - \lambda\left||F_m(\mathbf{z})| - k_m^*\right|. \tag{7}$$

The legacy-compatible mode ranks chromosomes by

$$\Phi_m^{\mathrm{legacy}}(\mathbf{z};\xi) = \max\{0, Q_m(\mathbf{z};\xi)\}. \tag{8}$$

The corrected-objective mode uses Q_m directly for reporting, ranking, elite selection, and run-best updating. Within one evaluated population, write $Q_i = Q_m(\mathbf{z}_i; \xi_i)$ and, when at least one objective is finite, $Q_{\min} = \min_{j:Q_j \text{ finite}} Q_j$. For proportional parent sampling only, nonnegative weights are obtained without changing the order of finite objectives:

$$w_i = \begin{cases} Q_i - Q_{\min} + \epsilon, & Q_i \text{ finite}, \\ 0, & \text{otherwise}, \end{cases} \tag{9}$$

where $\epsilon = 10^{-12}$. When the weights are valid, parent i is sampled with replacement using

$$\pi_i = \frac{w_i}{\sum_j w_j}. \tag{10}$$

Uniform sampling, $\pi_i = 1/N$, is recorded when no objective is finite, the finite-objective range is at most ϵ, or the weight total is invalid or effectively zero. Empty chromosomes are prohibited. The package records zero-truncation, target-deviation, sampling-fallback, objective-diversity, and mask-diversity diagnostics.

4.4 Regret-constrained medoid locking

For nonnegative absolute tolerance δ, the strict eligible pool is

$$E_\delta = \{r : R_r \le \delta\}. \tag{11}$$

It is never expanded with a candidate outside the declared threshold. Because a highest-scoring candidate has zero regret, E_δ is nonempty. If $|E_\delta| = 1$, its sole candidate is selected. Otherwise, each eligible candidate's mean Jaccard similarity is

$$\overline{J}_r = \frac{1}{|E_\delta| - 1} \sum_{\substack{s \in E_\delta \\ s \ne r}} \frac{|F_r \cap F_s|}{|F_r \cup F_s|}, \tag{12}$$

For $|E_\delta| > 1$, eligible candidates are ordered by four criteria, applied in sequence: larger mean Jaccard similarity, larger locking score, fewer selected features, and finally the lexicographically smaller stable mask hash. Let $\widehat{r}$ be the first candidate in that ordering. The locked feature set is

$$F^{\text{lock}} = F_{\widehat{r}}. \tag{13}$$

Here the stable canonical-mask hash is only a deterministic final tie-breaker, not a probabilistic or scientific score. Thus the selected candidate is the eligible-pool Jaccard medoid, subject to the stated score, feature-count, and hash tie-breakers. Exact duplicate candidates tied under this ordering have the same scientific feature set, retain their multiplicity as voting candidates, and use source-run provenance only after scientific feature-set selection.

Selection is restricted to E_δ, so the output always satisfies $R_{\widehat{r}} \le \delta$. At $\delta = 0$, all and only highest-scoring candidates are eligible; tied maxima still undergo medoid selection. If $\delta \ge \max_r R_r$, the rule becomes the full-bank medoid. These guarantees concern the configured empirical development-CV score gap, not expected predictive risk, generalization error, or statistical uncertainty. Complete propositions, proofs, duplicate-policy details, and implementation correspondence are given in the Supplementary Methods and Supplementary Table S25.

With R retained candidates, E eligible candidates, canonical-universe size p, and average selected-set size s, the package's dense-mask, sparse-Jaccard implementation has expected time $O(Rp+E^2 s+R \log R)$ and memory $O(Rp+E^2)$. For the present $R = 5$ design, at most 10 unordered eligible pairs are compared; this use case does not establish large-bank scalability. Detailed representation-specific derivations are in the Supplementary Methods.

The corrected-objective configuration fixes absolute $\delta = 0.01$ before locking. It is a metric-scale tolerance, not a universal threshold. Development-only sensitivity evaluates 0, 0.005, 0.01, 0.02, and a best-run-SE-scaled threshold where fold vectors exist. The scaled sensitivity is not the conventional paired one-standard-error rule and stops when required fold vectors are absent.

4.5 Controlled candidate-bank locking simulation

The frozen simulation protocol evaluated highest-locking-score selection, the legacy top-three medoid, the unrestricted full-bank Jaccard medoid, and the regret-constrained medoid under hidden synthetic utility. The primary design crossed three candidate topologies, three heterogeneity levels, three proxy strengths, and six score-noise ratios at $R = 5$ and $\delta = 0.01$, with 10,000 independently indexed banks per scenario. Prespecified sensitivities varied R, δ, score-noise distribution, and duplicate rate, yielding 513 scenarios and 4,995,000 banks in total. Primary summaries weight the 162 scenarios equally rather than treating their grid as a population prior. Oracle-regret failure was defined as hidden-utility regret greater than 0.02. The simulation used no participant-level data, classifier fitting, RFECV, Direct selection, or GA. Complete utility, candidate-generation, tie-path, execution-integrity, and claim-boundary details are provided in the Supplementary Methods, Supplementary Fig. S28, and Supplementary Table S23.

4.6 Development-only objective comparison

The comparison covered four AMP-AD held-out-center development partitions, three Direct branches, two feature caps, and five seeds per condition (24 conditions; 120 full GA runs). The matched original configuration used the archived zero-truncated objective and legacy top-three medoid. The corrected configuration used $\lambda = 0.015$, untruncated ranking and elitism, shifted parent-sampling weights, fixed five-fold development-CV macro one-vs-rest AUROC rescoring, and regret-constrained locking at $\delta = 0.01$. Because candidate generation and locking both changed, this was a configuration-level comparison. Held-out-center outcomes were not accessed.

4.7 One-time post-freeze AMP-AD held-out evaluation

All 24 masks, locking scores, hyperparameters, metric orientation, preprocessing boundary, final estimator, bootstrap plan, and source hashes were frozen in a SHA-256-identified protocol before guarded one-time evaluation at the corresponding held-out center. Each signature was fitted on its complete development partition using the frozen 150-tree random-forest specification; no result triggered reselection or configuration change. Macro one-vs-rest AUROC was primary, with macro AUPRC, balanced accuracy, macro F1, and accuracy secondary. Comparisons with RFECV-only and legacy-medoid predictions used 2,000 common resamples stratified by center and class (seed 42). The four centers represent four distribution shifts; the 24 conditions are not independent cohorts.

4.8 CGGA coherent benchmark and saved-bank nested sensitivity

The coherent benchmark reconstructed the archived 214/92 CGGA split and verified participant IDs, labels, and compact-matrix values against the raw source. RFECV-only, size-matched Elastic Net [22], highest-score, legacy-medoid, unrestricted-medoid, and regret-constrained-medoid signatures were evaluated with the same fixed 500-tree random forest. AUROC intervals and paired contrasts used 2,000 common class-stratified resamples (seed 42). No feature-selection stage was rerun.

For the nested sensitivity, the five saved candidate banks from a reduced-budget CGGA analysis were re-locked at $\delta = 0.01$ while preserving outer-fold preprocessing, feature universes, masks, and candidate scores. A 500-tree random forest was refitted in each archived outer-training fold. This post hoc internal sensitivity used 214 participants, five outer folds, and a historical 20-individual, 12-generation GA budget; it did not access the 92-sample held-out partition or replace the primary analysis.

4.9 Fully nested TCGA GBM/LGG analysis

The TCGA analysis used a supplied workbook containing 491 participants, 20,118 processed RNA-seq features, and 5,000 cleaned SCNV features. The provider confirmed label-independent variance filtering; a label-blind audit recovered 4,999 features by ranking the 24,776-feature uncleaned sheet, with an exact variance tie at the cutoff. This upstream reduction was the declared empirical starting boundary. Untreated primary GBM, astrocytoma, oligodendroglioma, and oligoastrocytoma counts were 66, 147, 165, and 113 [23–25]. Modality prefixes were retained; clinical columns, RPPA, and uncleaned SCNV were excluded.

Two shuffled stratified five-fold outer-CV repeats used random states 20260810 and 20260811. Within every outer-training partition, median imputation, zero-variance filtering, Direct screening, RFECV target estimation, five GA runs, fixed candidate rescoring, and locking were refitted without outer-test access. Post-Direct universes were capped deterministically at 100 features. RFECV, GA fitness, and locking used macro one-vs-rest AUROC. Each GA used 50 individuals, 50 generations, seeds 42–46, the corrected untruncated shifted-linear objective, and $\lambda = 0.015$. The five saved run-best candidates were rescored using a fixed five-fold, 500-tree random-forest evaluator and locked with absolute $\delta = 0.01$, strict eligible-only pooling, and canonical deterministic tie-breaking. This produced 30 outer-training candidate banks and 150 GA runs.

After locking, Direct, RFECV-only, highest-locking-score, legacy top-three-medoid, unrestricted-full-bank-medoid, and regret-constrained-medoid signatures were evaluated on the corresponding outer-test fold using the same 500-tree random forest and a common deterministic seed within each repeat–fold–branch condition. Mean repeat-specific OOF macro one-vs-rest AUROC was the primary predictive summary. Confidence intervals and paired regret-constrained-medoid-minus-comparator differences used 2,000 participant-clustered, class-stratified bootstrap resamples that retained both repeat-specific predictions for each sampled participant. The pairwise outer-fold Jaccard values share training observations and were therefore summarized descriptively rather than treated as independent replicates. Complete execution hashes, agreement definitions, duplicate handling, and source-table correspondence are reported in the Supplementary Methods.

The protocol-frozen selector-layer sensitivity reused each verified post-Direct outer-training universe without rerunning prior selection stages. Nested multinomial Elastic Net produced native and matched-cardinality supports; subsampling stability selection used 50 class-stratified half-samples and a 0.80 frequency threshold. Five random five-mask banks per condition matched feature count and locking-evaluation budget. Feature hashes preceded outer-test access, and signatures shared the condition-specific final-estimator seed. Because TCGA outcomes had already been analyzed, this is a retrospective nested sensitivity. Complete grids, seeds, integrity records, and reporting boundaries are in the Supplementary Methods and Supplementary Table S6.

4.10 Software implementation and audit artifacts

WrapEvoFS v0.2.0 supports Python 3.10–3.12 through an API and command-line interface. Configurations select legacy-compatible or corrected-objective search and locking modes; archived results are never silently replaced. Machine-readable outputs record scores, regrets, eligibility, overlap, selected masks, seeds, tie paths, fingerprints, and GA diagnostics. Pickle-free atomic checkpoints reject incompatible state. Complete configuration, schema, hashing, checkpoint, and serialization details are provided in the Supplementary Methods and repository documentation.

4.11 Empirical datasets and analysis boundary

The five empirical designs are summarized in Table 1, with preprocessing boundaries and archived original settings detailed in Supplementary Tables S26–S27. The ADNI-derived multiclass analysis contains 532 development and 133 held-out participants with 7,411 processed predictors [26, 27]. AMP-AD contains 527 participants and 13,136 supplied frontal-proteomic and metabolomic predictors from the AMP-AD Diverse Cohorts Study in four fixed leave-one-center-out evaluations [28]. The CGGA MGMT-methylation analysis contains 214 development and 92 held-out samples with 24,326 aligned gene-expression predictors [29]. The fully nested TCGA GBM/LGG analysis contains 491 participants with 20,118 supplied RNA-seq and 5,000 cleaned SCNV predictors [23, 24]. The VGH brain-tumour radiomics analysis contains 137 development and 60 held-out participants with 1,781 full-space features and a 1,346-feature perturbation-filtered sensitivity space. These are internal demonstrations beginning from supplied matrices; independent external validation was not performed.

The ADNI-derived source data were obtained from the Alzheimer's Disease Neuroimaging Initiative database (`https://adni.loni.usc.edu/`). ADNI was launched in 2003 as a public–private partnership led by Principal Investigator Michael W. Weiner, MD, to evaluate whether serial imaging, biological markers, and clinical and neuropsychological assessment can jointly measure progression of mild cognitive impairment and early Alzheimer's disease.

For the VGH brain-tumour radiomics analysis, prespecified header and geometry rules selected one eligible T1ce series per participant without MGMT access. DICOM series were converted to pseudonymized NIfTI with `dcm2niix` [30]; RaidionicsRADS and RaidionicsSeg generated automatic brain and tumor-core candidate regions [31], which were not manually corrected or treated as ground truth. MGMT linkage followed series selection, conversion, segmentation, configuration freezing, PyRadiomics extraction [32], and label-independent filtering. Of 326 source participants, 262 were T1ce eligible, 259 completed extraction, and 197 had unique validated MGMT linkage (117 unmethylated and 80 methylated); no labels were inferred for unlinked records. A fixed stratified 70:30 split yielded 137 development and 60 held-out participants. A separate deterministic 15-participant pilot froze the perturbation-filtered space using original, 1-mm-eroded, and 1-mm-dilated masks; retention required ICC(2,1) ≥ 0.85 and median relative range ≤ 0.25 without outcome labels [33]. Supplementary Table S28 reports the frozen extraction settings, complete source flow, and validation boundary; IBSI provides the feature-definition reference framework [34].

Within each feature space, SVM-L1, XGBoost, and Boruta-RF Direct screening and RFECV target estimation used development data only. Five GA searches used seeds 42–46, 50 individuals, 50 generations, five-fold ROC-AUROC scoring, the untruncated shifted-linear objective, and $\lambda = 0.015$. The five run-best sets were rescored by fixed five-fold development-CV ROC AUROC and locked with absolute $\delta = 0.01$, strict eligible-only pooling, and canonical deterministic tie-breaking. All six locks were completed before a 500-tree random forest was fitted on the full development partition and evaluated once on held-out data. AUROC and AUPRC intervals and paired AUROC differences used 2,000 class-stratified bootstrap resamples with seed 42. The two feature spaces share participants, split, and outcomes; their contrast is a feature-definition sensitivity analysis, not external validation.

Retrospective sensitivities used only available masks, scores, and fold vectors; missing candidate artifacts were not reconstructed. Complete cohort-specific availability is reported in the Supplement.

Except in the fully nested TCGA experiment, preprocessing and label-dependent Direct screening were fitted once on each full development partition rather than refitted inside every locking-score fold. RFECV, GA evaluation, and locking reused configured development folds during adaptive selection; the locking evaluator used fixed stratified five-fold construction with random state 42. Candidate evaluators used run-derived estimator seeds, so seed agreement may reflect search and estimator randomness. Development-CV scores are therefore adaptive internal selection scores rather than unbiased generalization estimates. In TCGA, the entire selection pipeline was refitted inside each outer-training partition, and only the resulting outer-test predictions contributed to repeated OOF performance summaries.

4.12 Stage-metric audit and secondary analyses

Archived configurations used mixed stage metrics: ADNI and AMP-AD combined weighted multiclass AUROC, balanced accuracy, and the locking metric; CGGA combined RFECV AUROC, GA accuracy, and locking AUROC. These original workflows are staged heuristics, not unified optimizers. The new binary radiomics analysis aligned RFECV, GA, and locking on ROC AUROC under the corrected-objective configuration. The package's `scoring.unified_metric: macro_ovr_auroc` option maps to binary ROC AUROC or multiclass macro one-vs-rest AUROC when compatible; it does not retroactively alter archived runs.

Held-out results were joined only after each development-only rule was fixed. A held-out estimate for a newly selected rule was reused only when its selected run exactly matched an archived evaluated run; otherwise it remains unavailable. The CGGA paired unrestricted-full-bank-medoid-minus-RFECV-only contrasts used 2,000 common class-stratified bootstrap resamples (seed 42) from the frozen aligned held-out predictions. These post hoc intervals did not affect candidate generation, locking, or the choice of δ; no held-out value compared locking strategies during selection.

Bayesian uncertainty quantification, training-only STABL, and BLiP are secondary interoperability analyses [35–39]. They show that development-derived locked outputs can define a downstream candidate set for posterior inference, resampling-oriented selection, and expected-FDR decisions. Detailed priors, diagnostics, stability paths, posterior summaries, and BLiP sensitivities are provided in the Supplementary Information. Cross-method overlap is not interpreted as biomarker validation.

4.13 Ethics and consent

This study performed secondary methodological analyses of existing data and involved no new participant recruitment, intervention, or recontact. ADNI protocols were approved by the institutional review board or research ethics board at each participating site, and written informed consent was obtained from participants or their authorized representatives. The CGGA source collection was approved by the Beijing Tiantan Hospital Institutional Review Board, was conducted in accordance with the Declaration of Helsinki, and obtained written informed consent; the CGGA resource identifies collection protocol KY2013-017-01. The AMP-AD analysis used de-identified post-mortem multi-omics data from the AMP-AD Diverse Cohorts Study contributed by Emory University, Mayo Clinic, Mount Sinai, and Rush University and accessed as supplied analysis matrices. The source study reports approval by the institutional review board at each respective institution and consent from all participants or next of kin [28]; the supplied matrices did not retain a single cross-cohort protocol identifier, and none is asserted here.

The TCGA GBM/LGG analysis used de-identified processed genomic matrices from an existing public research resource and involved no participant recontact. The source TCGA studies describe their collection and genomic characterization procedures [23, 24]. The historical histological labels used here are an analysis endpoint and are not asserted to represent current WHO integrated diagnoses.

The VGH brain-tumour radiomics analysis used existing clinical DICOM images and MGMT metadata under University of British Columbia research ethics approval H20-02354 (“Brain Tumor Image Analysis”; principal investigator Stephen Yip). The requirement for informed consent was waived for this retrospective study, and secondary analysis was permitted. DICOM data were de-identified during restricted preprocessing before radiomics analysis; direct identifiers remained confined to restricted linkage records, and participant-level data are not redistributed.

4.14 Use of generative AI

OpenAI's ChatGPT was used for coding assistance and grammar editing. The authors reviewed and verified all resulting changes and take full responsibility for the submitted work.

5 Data availability

Participant-level data are not redistributed; access requires the applicable source procedures. ADNI data are available to approved investigators through the LONI Image and Data Archive at `https://ida.loni.usc.edu/login.jsp?project=ADNI&page=HOME`. AMP-AD data are available through the AD Knowledge Portal at `https://adknowledgeportal.synapse.org/DataAccess`; the Diverse Cohorts Study is `syn51732482` (`https://doi.org/10.7303/9618093`). CGGA `mRNAseq_325` data are available at `https://www.cgga.org.cn/download.jsp`, and TCGA GBM and lower-grade-glioma source data through the NCI Genomic Data Commons at `https://portal.gdc.cancer.gov/`. The TCGA analysis used a provider-organized processed workbook whose redistribution permission has not been established; nonrestricted checksums and provenance manifests are supplied in the reproducibility repository. The VGH brain-tumour radiomics cohort cannot redistribute participant-level matrices, identifiers, images, masks, or prediction records; only non-identifying aggregate and audit materials are shared.

6 Code availability

WrapEvoFS v0.2.0 is publicly available under the BSD 3-Clause License at `https://github.com/junseoparkX/wrapevofs-package`; the tagged release is available at `https://github.com/junseoparkX/wrapevofs-package/releases/tag/v0.2.0`. The verified wheel and source distribution are available from `https://pypi.org/project/wrapevofs/0.2.0/`; install with `pip install wrapevofs==0.2.0`. Nonrestricted manuscript source, analysis code, figure source data, checksums, provenance manifests, and aggregate audit materials are publicly available at `https://github.com/junseoparkX/wrapevofs-manuscript-reproducibility`; restricted participant-level data are not included.

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Author contributions

J.P. performed conceptualization, methodology, software development, validation, formal analysis, investigation, data curation, DICOM de-identification, radiomics data generation, visualization, and writing of the original draft, and contributed to review and editing. L.J.F. provided supervision and contributed to review and editing. M.F.H. and E.S. provided the patient DICOM dataset and associated clinical metadata for the VGH brain-tumour radiomics cohort, supplied and verified the ethics and consent information used in the manuscript, and contributed to resources, data curation, and review and editing. H.Z. provided supervision and project administration, contributed to review and editing, and serves as corresponding author. All named authors reviewed and approved the manuscript and take responsibility for the work.

Competing interests

The authors declare no competing interests.